

R1



Gateway of last resort is not set

```
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      1.0.0.0/8 is directly connected, Ethernet0/1
L      1.0.0.1/32 is directly connected, Ethernet0/1
S      12.0.0.0/8 [1/0] via 172.50.0.2
S      29.0.0.0/8 [1/0] via 172.50.0.2
S      62.0.0.0/8 [1/0] via 172.50.0.2
S      129.23.0.0/16 [1/0] via 172.50.0.2
      172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.50.0.0/16 is directly connected, Ethernet0/0
L      172.50.0.1/32 is directly connected, Ethernet0/0
S      212.25.0.0/24 [1/0] via 172.50.0.2
S      222.215.0.0/24 [1/0] via 172.50.0.2
```

R1#

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#int e0/1

R1(config-if)#ip helper-address 172.50.0.2

R1(config-if)#

R1(config-if)#exit

R1(config)#

R1(config)#

R1(config)#exit

R1#

R1#

R1#

*Sep 7 10:30:40.615: %SYS-5-CONFIG_I: Configured from console by console

R1#

R3



a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 12.0.0.1
S    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
S    29.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    29.0.0.0/8 is directly connected, Ethernet0/1
L    29.0.0.1/32 is directly connected, Ethernet0/1
S    62.0.0.0/8 [1/0] via 12.0.0.1
S    129.23.0.0/16 [1/0] via 12.0.0.1
S    172.50.0.0/16 [1/0] via 12.0.0.1
S    212.25.0.0/24 [1/0] via 12.0.0.1
S    222.215.0.0/24 [1/0] via 12.0.0.1
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int e0/1
R3(config-if)#ip helper-address 222.215.0.1
R3(config-if)#exit
R3(config)#int e0/1
R3(config-if)#no ip helper-address 222.215.0.1
R3(config-if)#exit
R3(config)#int e0/1
R3(config-if)#ip helper-address 12.0.0.1
R3(config-if)#exit
R3(config)#
```

+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 222.215.0.1
S    12.0.0.0/8 [1/0] via 222.215.0.1
S    29.0.0.0/8 [1/0] via 222.215.0.1
S    62.0.0.0/8 [1/0] via 222.215.0.1
    129.23.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      129.23.0.0/16 is directly connected, Ethernet0/1
L      129.23.0.1/32 is directly connected, Ethernet0/1
S    172.50.0.0/16 [1/0] via 222.215.0.1
S    212.25.0.0/24 [1/0] via 222.215.0.1
    222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      222.215.0.0/24 is directly connected, Ethernet0/0
L      222.215.0.2/32 is directly connected, Ethernet0/0
```

R4#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R4(config)#

R4(config)#int e0/1

R4(config-if)#ip helper-address 62.0.0.1

R4(config-if)#exit

R4(config)#int e0/1

R4(config-if)#no ip helper-address 62.0.0.1

R4(config-if)#

R4(config-if)#int e0/1

R4(config-if)#ip helper-address 222.215.0.1

R4(config-if)#exit

R4(config)#

R4(config)#

R4(config)#exit

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 62.0.0.1
S    12.0.0.0/8 [1/0] via 62.0.0.1
S    29.0.0.0/8 [1/0] via 62.0.0.1
    62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      62.0.0.0/8 is directly connected, Ethernet0/0
L      62.0.0.2/32 is directly connected, Ethernet0/0
S    129.23.0.0/16 [1/0] via 62.0.0.1
S    172.50.0.0/16 [1/0] via 62.0.0.1
    212.25.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      212.25.0.0/24 is directly connected, Ethernet0/1
L      212.25.0.1/32 is directly connected, Ethernet0/1
S    222.215.0.0/24 [1/0] via 62.0.0.1
```

R5#

R5#

R5#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#int e0/1

R5(config-if)#ip helper-address 62.0.0.1

R5(config-if)#exi

R5(config)#

R2



N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 172.50.0.1
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/1
L    12.0.0.1/32 is directly connected, Ethernet0/1
S    29.0.0.0/8 [1/0] via 12.0.0.2
    62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    62.0.0.0/8 is directly connected, Ethernet0/3
L    62.0.0.1/32 is directly connected, Ethernet0/3
S    129.23.0.0/16 [1/0] via 222.215.0.2
    172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.50.0.0/16 is directly connected, Ethernet0/0
L    172.50.0.2/32 is directly connected, Ethernet0/0
S    212.25.0.0/24 [1/0] via 62.0.0.2
    222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    222.215.0.0/24 is directly connected, Ethernet0/2
L    222.215.0.1/32 is directly connected, Ethernet0/2
```

R2#

R2#

R2#

R2#conf t

```
R2
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp pool Sales
R2(dhcp-config)#network 1.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 1.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 1.0.0.1 1.0.0.100
R2(config)#exit
R2#
R2#
R2#
*Sep  7 10:25:34.599: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp pool HR
R2(dhcp-config)#network 29.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 29.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 29.0.0.1 29.0.0.100
R2(config)#exit
R2#
*Sep  7 10:27:01.113: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp pool Management
R2(dhcp-config)#network 129.23.0.0 255.255.0.0
R2(dhcp-config)#default-router 129.23.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
```

```
R2
R2(config)#ip dhcp pool Management
R2(dhcp-config)#network 129.23.0.0 255.255.0.0
R2(dhcp-config)#default-router 129.23.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 129.23.0.1 129.23.0.100
R2(config)#exit
R2#
R2#
*Sep  7 10:28:23.611: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip dhcp pool Prod
R2(dhcp-config)#network 212.25.0.0 255.255.255.0
R2(dhcp-config)#default-router 212.25.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 212.25.0.1 212.25.0.100
R2(config)#exit
R2#
R2#
R2#
R2#
*Sep  7 10:29:42.053: %SYS-5-CONFIG_I: Configured from console by console
R2#ip dhcp binding
^
% Invalid input detected at '^' marker.

R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip dhcp binding
```

R2#ip dhcp binding

^
% Invalid input detected at '^' marker.

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#ip dhcp binding

% Incomplete command.

R2(config)#exit

R2#sh

*Sep 7 10:36:36.218: %SYS-5-CONFIG_I: Configured from console by console

R2#show ip dhcp binding

Bindings from all pools not associated with VRF:

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
1.0.0.101	0150.b7bb.06ac.00	Sep 08 2026 04:01 PM	Automatic
29.0.0.101	0150.cd1b.06ad.00	Sep 08 2026 04:01 PM	Automatic
129.23.0.101	0150.ce12.06ae.00	Sep 08 2026 04:03 PM	Automatic

R2# show ip dhcp binding

Bindings from all pools not associated with VRF:

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
1.0.0.101	0150.b7bb.06ac.00	Sep 08 2026 04:01 PM	Automatic
29.0.0.101	0150.cd1b.06ad.00	Sep 08 2026 04:01 PM	Automatic
129.23.0.101	0150.ce12.06ae.00	Sep 08 2026 04:03 PM	Automatic
212.25.0.101	0150.6531.06af.00	Sep 08 2026 04:08 PM	Automatic

R2#







